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Roll No

EE-7203 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Generalized Theory of Electrical Machines
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. What is connection matrix? Explain its use in obtaining transformed or new impedance and voltage matrix.
2. Draw and explain torque-slip curve for a polyphase induction machine in braking, motoring and generating region.
3. Discuss the concept of transformation change of variables and machine variables and transform variables, their application to DC machine for steady state analysis.
4. Write short notes on any two of the following:
 - a) Equation of cross field commutator machine
 - b) The problem of power system analysis
 - c) Standard synchronous machine reactances.
5. Deduce park's transformations relating the 3-phase currents of a synchronous machine to its corresponding d - q axes currents.
6. Discuss the parameters from short circuit characteristics.
7. Explain application of approximate methods to power system analysis.

OR

Obtain an expression for electrical torque of Kron's primitive machine. Also show that no torque is produced by interaction between the flux and current on the same axis.

8. Describe linear transformation in machine from three phases to two phases (a, b, c to α , β , O)
